PlatformIO Setup Guide

This guide provides instructions for setting up PlatformIO for developing and managing your project.

Prerequisites

Before starting, ensure you have the following installed on your system:

- Visual Studio Code (VS Code): https://code.visualstudio.com/

- Python (version 3.6 or later)

- PlatformIO Core CLI (optional but recommended for advanced users)

Installation Steps

1. Install PlatformIO Extension for VS Code

Open VS Code.
Go to the **Extensions** view (Ctrl+Shift+X or Cmd+Shift+X on macOS).
Search for 'PlatformIO IDE' and click Install.

2. Verify Installation

Open the PlatformIO Home screen by clicking on the PlatformIO icon in the Activity Bar.
Ensure PlatformIO initializes correctly and loads its dashboard.

3. Open an Existing PlatformIO Project (Platformio-Firmware)

To open an existing project:
1. Go to **PlatformIO Home > Open Project**.
2. Navigate to the folder containing your PlatformIO project.
3. Select the folder and click Open.
Alternatively, you can use **File > Open Folder** in VS Code and select your project folder.

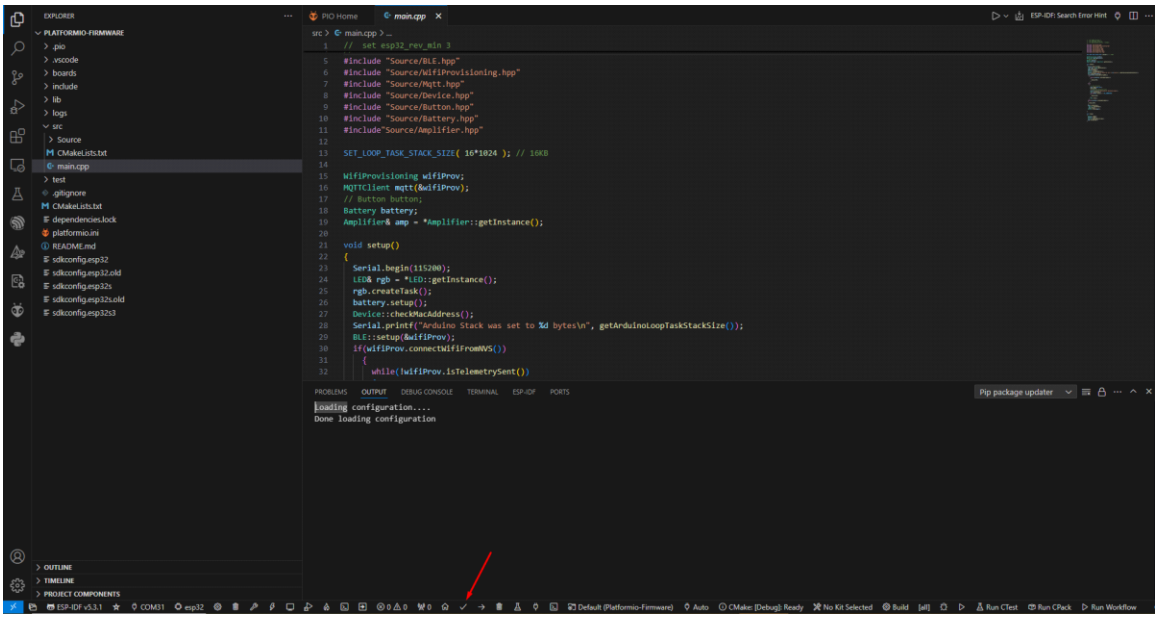
4. Configure the Existing `platformio.ini` File

Open the `platformio.ini` file located in the root directory of your project.
This file is already created when you set up your project and contains default settings for your selected board and framework.
Press (Ctrl + S) to download the packages and libraries. After Download and configuration.

Building and Uploading Code

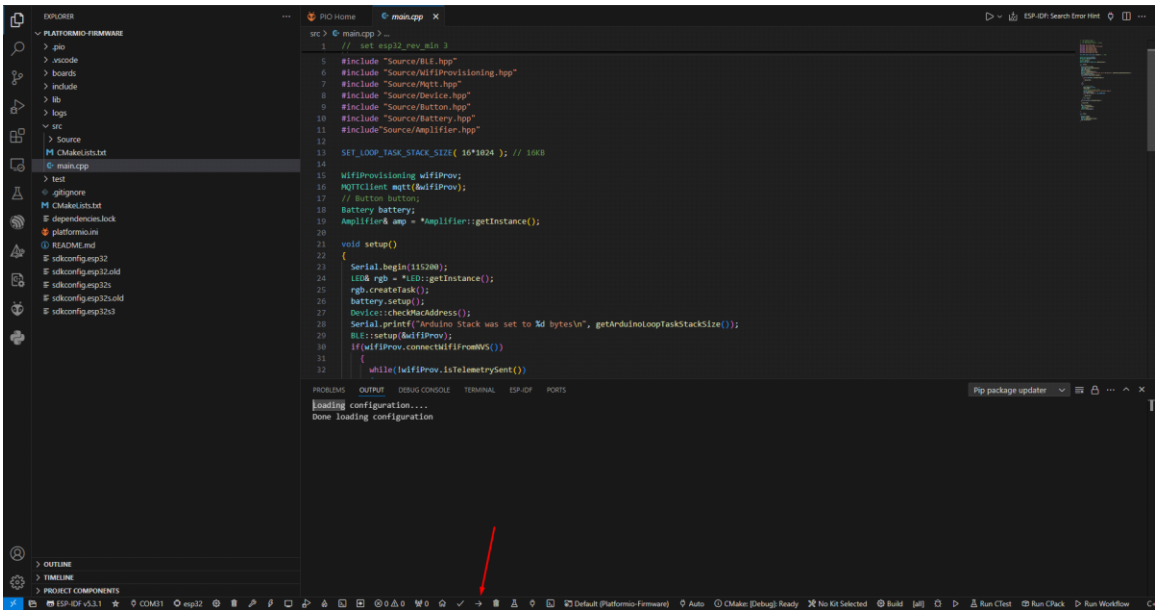
1. Build the Project

Click on the checkmark icon in the PlatformIO toolbar:



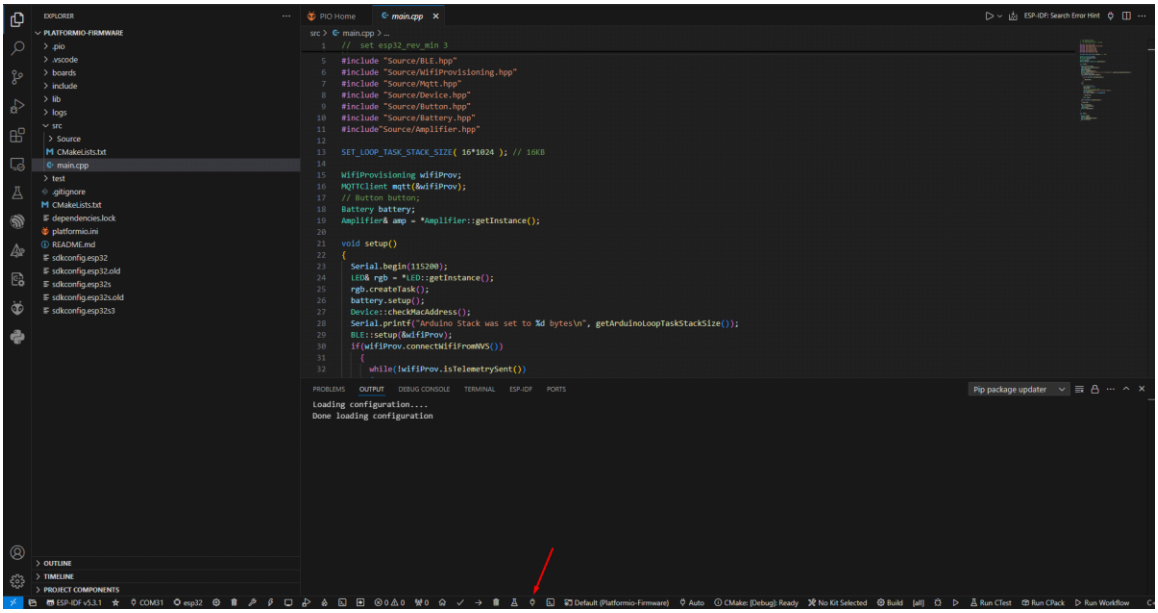
2. Upload Code to the Board

Click on the arrow icon in the PlatformIO toolbar:



3. Monitor Serial Output

Use the plug icon in the PlatformIO toolbar:



Common Issues and Troubleshooting

Detected a whitespace character in project paths

Ensure your project path does not contain spaces. Move the project to a location without spaces, e.g., C:\Projects\MyProject.

Board Not Detected

Check that the correct port is selected in the `platformio.ini` file.
Example:

"upload_port = COM3"